Water Quality Differences Between Freshwater Wetlands with Various Influences

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INTRODUCTION

- **Hydrology** is crucial in shaping the physiochemical and ecological conditions of wetlands, influencing overall health and function (Deemy et al. 2022).
- **Stressors** like anthropogenic impacts, vegetation changes, and soil dynamics can significantly affect wetland ecosystems (Herlihy et al. 2019).
- **Water quality** indicators, such as pH, temperature, dissolved oxygen, turbidity, and conductivity, determine the health of wetland ecosystems (Guidugli-Cook et al. 2017; Husband and McIntyre 2024).
- **Vegetation** acts as a natural filter in wetlands, enhancing water quality and supporting biodiversity; increased vegetation can improve dissolved oxygen levels (Morrison and Bohlen 2010).
- **Conservation** is crucial because healthy wetlands provide essential ecosystem services; with a global loss of about 50% of wetlands, monitoring water quality and vegetation growth is critical for conservation efforts (Mitsch and Gosselink 2007; Glicksman 2014).

STUDY SITES

OPES Site

GIP Site

CCGA Site

OBJECTIVES

The objectives of this study were to:

1. Quantify water quality parameters of three varied types of wetlands at weekly intervals.
2. Compare water quality parameters between the wetlands over multiple weeks.
3. Determine how increased urbanization affects water quality parameters.

METHODS

- Sampling occurred every 5 meters around the edge of each wetland until collecting 10 data points.
- Water quality parameters were measured using a YSI Proquatro multi-probe and a LaMotte turbidimeter and logged.
- Data from each sampling site was analyzed (pooled and averaged, median taken and averaged, etc.).
- An ANOVA test was performed to determine if data was significantly different .

RESULTS

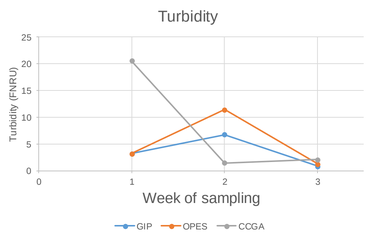


Figure 1. Median turbidity data for each site across a three-week timespan.

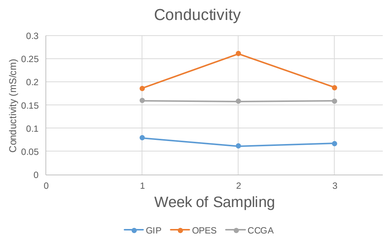


Figure 2. Median conductivity data for each site across a three-week timespan.

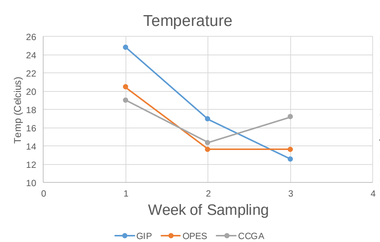


Figure 3. Median temperature data for each site across a three-week timespan.

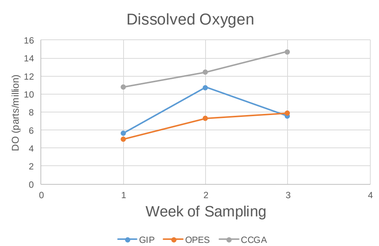


Figure 4. Median dissolved oxygen data for each site across a three-week timespan.

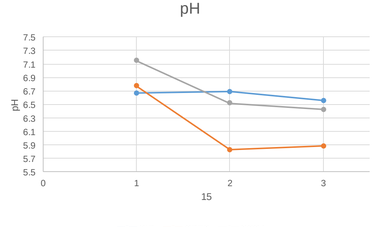


Figure 3. Median pH data for each site across a three-week timespan.

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	0.522	2.000	0.261	1.769	0.249	5.143
Within Groups	0.885	6.000	0.148			
Total	1.408	8.000				

Table 1. An ANOVA test on the median data of pH, a main determinant of urban pollution.

CONCLUSIONS

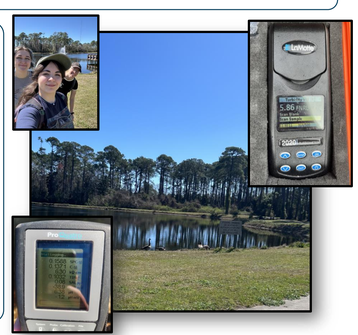
Given our results, we conclude that:

1. Water quality parameters stayed relatively similar across sites excluding:
   - Conductivity increased with decreasing urbanization, possibly due to increased dissolved solids in urban areas.
   - Dissolved oxygen was lowest at OPES, likely due to higher vegetation cover.
2. Most parameters stayed consistent over time.
   - Turbidity, temperature, and pH might have been impacted from precipitation during the last samples, leading to decreases over time.
3. Urbanization of wetlands has a significant difference in dissolved oxygen and conductivity based on ANOVA tests.

Future Objectives:

1. Collect a larger range of data to see long term impacts on our sites
2. Examine how leaf out effects the overall health of the wetlands as the season change.
3. Test nutrient parameters on the sites (sulfur, phosphorus, and nitrogen concentrations).

References



Acknowledgements

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